WoSM Academy

MAT231-F27 · Calculus II

Second derivatives

The Hessian and the multivariable Taylor expansion

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Last time

- The Jacobian matrix $DF(\mathbf{a})$: rows are the gradients of the components.
- **Chain rule**: $D(F \circ G)(\mathbf{a}) = DF(G(\mathbf{a}))\, DG(\mathbf{a})$.
- Backpropagation: the chain rule in reverse order gives the whole gradient in one pass.
- $|\det DF|$ is the local volume factor.

First derivatives give the best linear approximation: the tangent plane. Today: second derivatives, which describe how f *curves* away from it — the information we need to tell summits from pits and saddles (Lecture 14).

This lecture

01 Higher partial derivatives

02 Schwarz's theorem

03 The Hessian matrix

04 Taylor expansion in several variables

05 Quadratic forms and curvature

06 Higher-order Taylor expansions (optional)

07 Summary and practice

Reading: J. E. Marsden, A. Tromba, *Vector Calculus*, Sections 3.1 and 3.2.

01

Higher partial derivatives

Second and higher partial derivatives

Partial derivatives of partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{xy} = (f_x)_y = \frac{\partial^2 f}{\partial y\,\partial x}, \qquad f_{yx} = (f_y)_x = \frac{\partial^2 f}{\partial x\,\partial y}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2}.$$

- In the subscript notation the order of differentiation is read from left to right; in the ∂ notation, from right to left.
- A function of n variables has n^2 second partial derivatives; third derivatives such as f_{xyz} are formed the same way.

Smoothness classes

Definition

f is of class C^k on an open set U if all its partial derivatives of order $\leq k$ exist and are continuous on U; it is C^∞ if this holds for every k.

- Polynomials, e^x, sin, cos and their sums, products, quotients and compositions are C^∞ where defined.
- $x|x|$ is C^1 but not C^2 at 0: its derivative $2|x|$ has a corner.
- C^1 functions are differentiable (Lecture 10); C^2 functions have a second-order Taylor expansion (this lecture).

Examples

Example ($f(x,y) = x^3y^2 + e^{xy}$)

$$f_x = 3x^2y^2 + y\,e^{xy}, \quad f_y = 2x^3y + x\,e^{xy},$$

$$f_{xx} = 6xy^2 + y^2e^{xy}, \qquad f_{yy} = 2x^3 + x^2e^{xy},$$
$$f_{xy} = 6x^2y + (1+xy)\,e^{xy}, \qquad f_{yx} = 6x^2y + (1+xy)\,e^{xy}.$$

Example ($f(x,y) = \sin(x+2y)$)

$$f_{xx} = -\sin(x+2y), \quad f_{xy} = f_{yx} = -2\sin(x+2y), \quad f_{yy} = -4\sin(x+2y).$$

In both, $f_{xy} = f_{yx}$. That is no accident.

02

Schwarz's theorem

Mixed partial derivatives commute

Theorem (Schwarz, or Clairaut)

If f is C^2 near (a, b), then $f_{xy}(a, b) = f_{yx}(a, b)$.

Proof sketch: the mean value theorem twice.

Let $\Delta = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b)$.

- With $\varphi(x) = f(x, b+k) - f(x, b)$:
  $\Delta = \varphi'(\xi)\, h = \big(f_x(\xi, b+k) - f_x(\xi, b)\big)h = f_{xy}(\xi, \eta)\, hk.$
- With $\psi(y) = f(a+h, y) - f(a, y)$, the same steps give $\Delta = f_{yx}(\xi', \eta')\, hk$.

Divide by hk, let $h, k \to 0$, and use continuity. □

The hypothesis matters

Example

$f(x,y) = \frac{xy(x^2-y^2)}{x^2+y^2}$ for $(x,y) \neq (0,0)$, $f(0,0) = 0$.

- $f_x(0,y) = \lim_{h\to 0} \frac{f(h,y)}{h} = -y$, so $f_{xy}(0,0) = -1$.
- $f_y(x,0) = \lim_{k\to 0} \frac{f(x,k)}{k} = x$, so $f_{yx}(0,0) = +1$.

Warning

f is C^1, and both mixed partial derivatives exist at $(0,0)$, but they are not continuous there — and they differ. Schwarz's theorem needs continuous second derivatives.

03

The Hessian matrix

The Hessian matrix

Definition

The **Hessian matrix** of $f\colon \mathbb{R}^n \to \mathbb{R}$ at $\mathbf{a}$ is the $n \times n$ matrix of second partial derivatives,

$$H_f(\mathbf{a}) = \left(f_{x_i x_j}(\mathbf{a})\right)_{i,j=1,\dots,n} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}(\mathbf{a}) \quad \text{for } n = 2.$$

- For C^2 functions, H_f is **symmetric** (Schwarz).
- H_f is the Jacobian of the gradient: $H_f(\mathbf{a}) = D(\nabla f)(\mathbf{a})$. Row i is the gradient of $\partial f/\partial x_i$.

Hessians: examples

Example ($f = x^3y^2 + e^{xy}$, from the start of the lecture)

$$H_f = \begin{pmatrix} 6xy^2 + y^2e^{xy} & 6x^2y + (1+xy)e^{xy} \\ 6x^2y + (1+xy)e^{xy} & 2x^3 + x^2e^{xy} \end{pmatrix}, \qquad H_f(1,0) = \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}.$$

Example (Three variables: $g = xyz + z^2$)

$$\nabla g = (yz,\ xz,\ xy + 2z), \text{ so } H_g = \begin{pmatrix} 0 & z & y \\ z & 0 & x \\ y & x & 2 \end{pmatrix}.$$

Quadratic functions

For a fixed $n \times n$ matrix A, vector $\mathbf{b}$ and number c, let $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$.

- $\nabla f(\mathbf{x}) = (A + A^\top)\mathbf{x} + \mathbf{b}$ (Lecture 11).
- $H_f(\mathbf{x}) = D\big((A + A^\top)\mathbf{x} + \mathbf{b}\big) = A + A^\top$: constant.
- With A symmetric, $H_f = 2A$. So in optimisation one often writes $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$ with Q symmetric: then $H_f = Q$.

Example

$f(x, y) = x^2 + 3xy + 2y^2 - x + 5$: $H_f = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$ everywhere.

Least squares (Lecture 11): $\|A\mathbf{x} - \mathbf{b}\|^2$ has Hessian $2A^\top A$.

04

Taylor expansion in several variables

Reduce to one variable

Fix $\mathbf{a}$ and $\mathbf{h}$, and look at f along the segment from $\mathbf{a}$ to $\mathbf{a}+\mathbf{h}$: $g(t) = f(\mathbf{a}+t\mathbf{h})$, $0 \le t \le 1$. By the chain rule (Lecture 12):

$$g'(t) = \nabla f(\mathbf{a}+t\mathbf{h}) \cdot \mathbf{h} = \sum_i f_{x_i}(\mathbf{a}+t\mathbf{h})\, h_i,$$

$$g''(t) = \sum_{i,j} f_{x_i x_j}(\mathbf{a}+t\mathbf{h})\, h_i h_j = \mathbf{h}^{\top} H_f(\mathbf{a}+t\mathbf{h})\, \mathbf{h}.$$

Taylor's theorem in one variable (Lecture 6), $g(1) = g(0) + g'(0) + \frac{1}{2}g''(\tau)$ for some $\tau \in (0,1)$, gives

$$f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \tfrac{1}{2}\, \mathbf{h}^{\top} H_f(\mathbf{a}+\tau\mathbf{h})\, \mathbf{h}.$$

The second-order Taylor expansion

Theorem

If f is C^2 near $\mathbf{a}$, then

$$f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a})\cdot\mathbf{h} + \tfrac{1}{2}\,\mathbf{h}^{\top} H_f(\mathbf{a})\,\mathbf{h} + o(\|\mathbf{h}\|^2).$$

Proof: replace $H_f(\mathbf{a}+\tau\mathbf{h})$ by $H_f(\mathbf{a})$; the error is at most $\frac{1}{2}\|H_f(\mathbf{a}+\tau\mathbf{h}) - H_f(\mathbf{a})\|\,\|\mathbf{h}\|^2$, and the first factor tends to 0 by continuity.

For $n = 2$, with $\mathbf{h} = (h, k)$ and all derivatives at (a, b):

$$f(a+h, b+k) \approx f + f_x h + f_y k + \tfrac{1}{2}\big(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2\big).$$

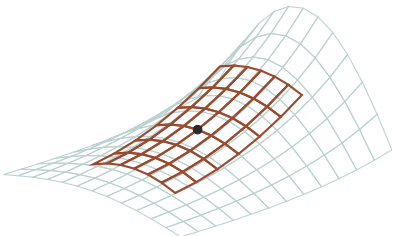
Constant, linear and quadratic terms: the tangent plane, corrected by a quadratic form.

Example: $e^x \cos y$ at $(0,0)$

At $(0,0)$: $f = 1$, $f_x = e^x \cos y = 1$,
$f_y = -e^x \sin y = 0$,
$f_{xx} = 1$, $f_{xy} = -e^x \sin y = 0$,
$f_{yy} = -e^x \cos y = -1$:

$$T_2(x,y) = 1 + x + \tfrac{1}{2}(x^2 - y^2).$$

Check with series:
$(1 + x + \frac{x^2}{2} + \cdots)(1 - \frac{y^2}{2} + \cdots)$.
At $(0.1, 0.2)$: $e^{0.1} \cos 0.2 = 1.08314$; the tangent plane gives 1.1, and T_2 gives 1.085.



The graph, and the quadratic surface $z = T_2$ touching it at $(0,0,1)$

Example: $\ln(1+x+2y)$ at $(0,0)$

With $u = 1+x+2y$:

$$\nabla f = \frac{1}{u}\begin{pmatrix}1\\2\end{pmatrix}, \qquad H_f = -\frac{1}{u^2}\begin{pmatrix}1 & 2\\2 & 4\end{pmatrix}.$$

At $(0,0)$, $u = 1$:

$$T_2(x,y) = 0 + (x+2y) - \tfrac{1}{2}(x^2+4xy+4y^2) = (x+2y) - \tfrac{1}{2}(x+2y)^2.$$

The same as $\ln(1+v) = v - \frac{v^2}{2} + \cdots$ with $v = x+2y$ (Lecture 6) — by uniqueness, any correct quadratic approximation is the Taylor polynomial. At $(0.1, 0.05)$: $\ln 1.2 = 0.18232$ and $T_2 = 0.18$.

05

Quadratic forms and curvature

The quadratic term is a quadratic form

$Q(\mathbf{h}) = \mathbf{h}^\top H \mathbf{h}$ with $H = H_f(\mathbf{a})$ symmetric is a **quadratic form** (Linear Algebra, Unit 21).

- Spectral theorem (Linear Algebra, Unit 20): $H = U\Lambda U^\top$ with U orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, the eigenvalues.
- In the rotated coordinates $\mathbf{u} = U^\top \mathbf{h}$ (along the eigenvectors):

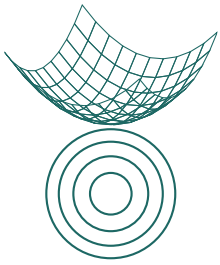
$$\mathbf{h}^\top H \mathbf{h} = \lambda_1 u_1^2 + \lambda_2 u_2^2 + \cdots + \lambda_n u_n^2.$$

At a point where $\nabla f(\mathbf{a}) = \mathbf{0}$:

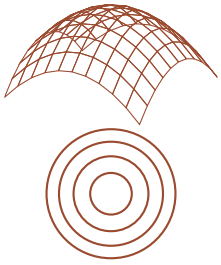
$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \tfrac{1}{2}(\lambda_1 u_1^2 + \cdots + \lambda_n u_n^2).$$

The eigenvectors are the **principal directions**, and the eigenvalues are the **principal curvatures** of the approximating quadratic surface: along u_i it is the parabola $\frac{1}{2}\lambda_i u_i^2$.

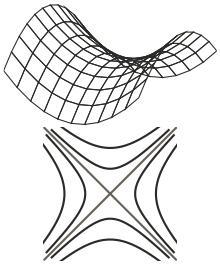
Three shapes



Bowl: $x^2 + y^2$
all $\lambda_i > 0$: positive definite



Cap: $-x^2 - y^2$
all $\lambda_i < 0$: negative definite



Saddle: $x^2 - y^2$
λ_i of both signs: indefinite

Examples

- $f = x^2 + 3xy + 2y^2$: $H = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$, eigenvalues $3 \pm \sqrt{10} \approx 6.16,\ -0.16$: a **saddle**, although every coefficient is positive.
- $f = 2x^2 + 2xy + 2y^2$: $H = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$, eigenvalues 6 and 2: a **bowl**. The principal directions are $(1, 1)$ and $(1, -1)$.

For 2×2 matrices

$\det H = \lambda_1 \lambda_2$ and $\operatorname{tr} H = \lambda_1 + \lambda_2$. So: $\det H < 0$ means a saddle; $\det H > 0$ means a bowl if $\operatorname{tr} H > 0$ and a cap if $\operatorname{tr} H < 0$. If $\det H = 0$, the quadratic term alone does not decide (Lecture 14).

06

Higher-order Taylor expansions (optional)

Multi-index notation

A **multi-index** is $\alpha = (\alpha_1, \dots, \alpha_n)$ with integers $\alpha_i \geq 0$. Write

$$|\alpha| = \alpha_1 + \cdots + \alpha_n, \quad \alpha! = \alpha_1! \cdots \alpha_n!, \quad \mathbf{h}^\alpha = h_1^{\alpha_1} \cdots h_n^{\alpha_n}, \quad \partial^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

Theorem

If f is C^k near $\mathbf{a}$, then $f(\mathbf{a}+\mathbf{h}) = \displaystyle\sum_{|\alpha| \leq k} \frac{\partial^\alpha f(\mathbf{a})}{\alpha!}\, \mathbf{h}^\alpha + o(\|\mathbf{h}\|^k)$.

Check for $|\alpha| = 2$, $n = 2$: $\alpha = (2,0), (1,1), (0,2)$ give $\frac{f_{xx}}{2} h^2 + f_{xy}\, hk + \frac{f_{yy}}{2} k^2$ — the quadratic term from before.

A third-order term

Example ($e^x \cos y$ at $(0,0)$, continued)

Third partial derivatives at $(0,0)$: $f_{xxx} = 1$, $f_{xxy} = 0$, $f_{xyy} = -1$, $f_{yyy} = 0$. The third-order terms:

$$\frac{f_{xxx}}{3!}x^3 + \frac{f_{xxy}}{2!\,1!}x^2y + \frac{f_{xyy}}{1!\,2!}xy^2 + \frac{f_{yyy}}{3!}y^3 = \frac{x^3}{6} - \frac{xy^2}{2}.$$

At $(0.1, 0.2)$: $T_3 = 1.083167$, against 1.083141.

- In n variables there are $\binom{n+k-1}{k}$ terms of order k. For $n = 1000$: $500\,500$ of order 2, and $167\,167\,000$ of order 3.
- So in several variables one rarely goes beyond second order. Optimisation methods (Lectures 14–15) use the gradient and, at most, the Hessian.

07

Summary and practice

Summary

- Second partial derivatives $f_{xx}, f_{xy}, \ldots$; the classes C^k.
- **Schwarz**: for C^2 functions $f_{xy} = f_{yx}$; without continuity it can fail.
- The **Hessian** H_f is symmetric for C^2 functions, and it is the Jacobian of ∇f. For $\mathbf{x}^\top A\mathbf{x} + \mathbf{b}^\top\mathbf{x} + c$: $H_f = A + A^\top$.
- **Taylor**: $f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a})\cdot\mathbf{h} + \frac{1}{2}\mathbf{h}^\top H_f(\mathbf{a})\mathbf{h} + o(\|\mathbf{h}\|^2)$, from the one-variable formula along $t \mapsto \mathbf{a} + t\mathbf{h}$.
- The quadratic term is a quadratic form: its eigenvalues are the principal curvatures. Bowl (positive definite), cap (negative definite), saddle (indefinite).
- Higher orders: multi-indices; the number of terms grows quickly.

Practice

1. Find all second partial derivatives of $f(x, y) = x^2y^3 - 4xy + y^4$, and check that $f_{xy} = f_{yx}$.
2. Find the Hessian of $f(x, y, z) = x^2 + y^2 + z^2 + xy + yz$.
3. Find the second-order Taylor polynomial of $\sqrt{1 + x + y}$ at $(0, 0)$.
4. Find the second-order Taylor polynomial of xe^y at $(1, 0)$.
5. Bowl, cap or saddle? $H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, $\begin{pmatrix} -3 & 1 \\ 1 & -1 \end{pmatrix}$.
6. Show that $H_f = D(\nabla f)$, and use it to find the Hessian of $f(\mathbf{x}) = \|\mathbf{x}\|^2$.
7. Find the third-order Taylor polynomial of $\sin(x + y)$ at $(0, 0)$.

Answers

1. $f_{xx} = 2y^3$, $f_{xy} = f_{yx} = 6xy^2 - 4$, $f_{yy} = 6x^2y + 12y^2$.
2. $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.
3. $1 + \frac{x+y}{2} - \frac{(x+y)^2}{8}$ (the binomial series with $v = x + y$).
4. $1 + (x-1) + y + (x-1)\,y + \frac{y^2}{2}$.
5. Bowl (eigenvalues $3, 1$); saddle $(3, -1)$; cap ($\det = 2 > 0$, $\operatorname{tr} = -4 < 0$).
6. Row i of $D(\nabla f)$ is $\nabla(f_{x_i})$, that is $(f_{x_i x_1}, \ldots, f_{x_i x_n})$. For $\|\mathbf{x}\|^2$: $\nabla f = 2\mathbf{x}$, so $H_f = 2I$.
7. $(x+y) - \frac{(x+y)^3}{6}$.

Next time

Lecture 14 — Unconstrained optimisation (Unit 14)

- Critical points, where $\nabla f = \mathbf{0}$
- The second-derivative test, using the Hessian
- Convex functions: every critical point is a global minimum

Reading: Strang & Herman, *Calculus, Volume 3*, Section 4.7; for convexity, Boyd & Vandenberghe, *Convex Optimization*, Section 3.1.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.